

Alexis Thomas

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Applied scientist with expertise in statistical modeling, time series forecasting, software engineering and cloud architectures.

WORK EXPERIENCE

Amazon

Applied Scientist II

London, UK

Sept. 2024 –

Present

- Built a **long-term demand forecasting system** for a 3-year planning horizon, combining optimization algorithms with statistical modeling to satisfy multiple business constraints across 10 marketplaces and 31 countries, achieving 99.7% constraint satisfaction while serving capacity planning and financial forecasting workflows.
- Developed a **short-term granular demand forecasting model** for 2-month operational horizons, implementing hierarchical neural networks and statistical time series models to predict demand at zip-code level, achieving an average of 2% WAPE.
- Led **comprehensive model evaluation** across statistical, machine learning and foundation model paradigms, developing adaptive forecasting strategies that automatically adjust to different volatility regimes and market conditions.

Software Development Engineer II

Paris, France

March. 2022 – Sept. 2024

- Led **software architecture, development, and engineering** within a research team providing long-term vision to accelerate European delivery speed and optimize the Amazon delivery network.
- Main projects:
 - Building a platform enabling one-click execution of operational research model workflows, achieving cost-effective **scalability** to handle thousands of concurrent executions without pre-provisioned servers.
 - Input validation, sanitization, and **infrastructure as code** solutions, reducing production time for models and enhancing maintenance efficiency.
 - Data visualization platform with hundreds of internal users, integrating automated reporting, data pipelines, and big data processing.
- **Stack:** Python, Java, PostgreSQL, and AWS services (EC2, Lambda, ECS, Fargate, Glue, S3, RDS, Redshift...).

BMW Group France

Project Manager Intern

Paris, France

Feb. 2021 – July. 2021

- Collaborated with project managers to frame projects and monitored progress to completion.
- Provided financial and human resources estimates and identified risks and opportunities for the BMW France project portfolio.

Amazon Web Services

Software Development Engineer Intern

Dublin, Ireland

June. 2020 – Dec.

2020

- Developed a new service allowing customers to adjust internal tool configurations in real-time.
- Led end-to-end project management, including scoping, design, implementation, and testing.
- Utilized Java & AWS for backend development and React for frontend design.

EDUCATION

Mines Paris - Engineering School

Master's degree with major in computer science (*Diplôme d'ingénieur civil des Mines de Paris*) Paris, France 2018 – 2022

One of the top three French engineering schools. Advanced courses in Mathematics and Computer Science.

Lycée Saint-Louis

2016 – 2018

Two-year prep school preparing the French national competitive exams to engineering schools.

PUBLICATIONS

- "Forecasting Electric Vehicle Charging Station Occupancy" – DMLR (3rd place, Smarter Mobility Challenge) <https://data.mlr.press/assets/pdf/v01-16.pdf>.

CERTIFICATIONS & INTERESTS

- AWS Certified Solutions Architect – Professional (2024), TOEFL (2022).
- Development of a mobile & web app called Giftruly using **Large Language Models**.
- Interests in Singing; working on pet projects; crossfit; gaming.

TECHNICAL SKILLS

- Languages & Tools: Python, Java, JavaScript/TypeScript, ReactJS
- ML/Statistics: Time series forecasting, optimization, statistical modeling
- Cloud & Infrastructure: AWS (EC2, Lambda, S3, RDS, Redshift, Glue), Docker, infrastructure as code
- Data & Analytics: PostgreSQL, Spark, data pipelines